Rhizophore

In many species long, cylindrical, unbranched and leafless structures arise from the lower side of the stem at the point of dichotomy called rhizophores. They grow vertically downwards and produce tufts of adventitious roots.

Stem

The stem may be erect, dichotomously branched or prostrate with lateral branching. The prostrate stem is dorsiventral.

Leaves

The leaves are microphyllous, sessile and simple. A single midvein is present in the leaves. The vegetative leaf as well as the sporophyll bears a small membranous tongue like structure on adaxial surface called **ligule**. The basal part of the ligule possess a hemispherical mass of thin walled cells called **glossopodium**. The function of ligule is not known, but it is viewed to be associated with water absorption, secretion and prevent dessication of shoot. The members belonging to homeophyllum type possess same type of leaves spirally arranged on the stem whereas the heterophyllum type have two types of leaves- two dorsal rows of small leaves(Microphylls) and two ventral rows of large leaves(Megaphylls).

Internal structure

Root

The transverse section of the root reveals an outermost layer called epidermis. It is made up of tangentially elongated cells. The cortex is homogeneous made up of thin walled parenchyma. The innermost layer of cortex is called endodermis. The stele is a protostele, monarch and xylem is exarch (Figure 2.17).

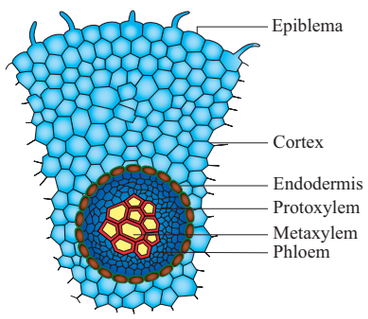


Figure 2.17: T.S. of root

Stem

The anatomy of the stem reveals the presence of epidermis, cortex and stelar region (Figure 2.18).

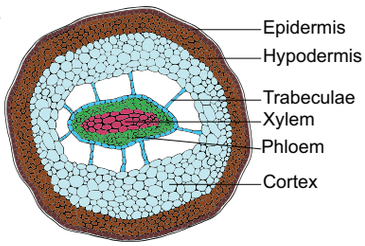


Figure 2.18: T.S. of Stem

The epidermis is parenchymatous and is covered with a thick cuticle. The cortex is parenchymatous with cells arranged without intercellular spaces. A sclerenchymatous hypodermis is noticed in *Selaginella lepidophylla*. The presence of radially elongated endodermal cells called **trabeculae** is the characteristic feature of *Selaginella*. The casparian strips are found on the lateral walls. The rapid stretching of the innermost cortical cells in comparison with stele results in air spaces and stele appears to be suspended in air space with the help of trabeculae. The stele is a protostele and exarch. A variation in number of steles is found. It may be monostelic (*S. spinulosa*); distelic (*S. kraussiana*) or polystelic (*S. laevigata*). The xylem is monarch(*S. kraussiana*) or diarch (*S. oregana*). Tracheids are present but vessels are also noticed in *S. densa* and *S. rupestris*.

Leaf

The leaf shows upper and lower epidermis. The epidermal cells have chloroplast. Stomata occur on both surfaces. The mesophyll is made up of loosely arranged thin walled cells with intercellular spaces. There is a median vascular bundle surrounded by a bundle sheath. In vascular bundle xylem is surrounded by phloem.

Reproduction

Vegetative reproduction

Selaginella reproduces vegetatively by fragmentation, bulbil formation, tuber formation and resting buds.

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